

# SUMMER INTERNSHIP PROJECT REPORT

## *Crop Yield Estimation Using Time Series Models*

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**Ankit Bhar**  
*3rd year, Bachelor of  
Statistics, ISI Kolkata*

**Konkana Adhikary**  
*3rd year, Bachelor of Science  
in Statistics, St. Xavier's  
College, Kolkata*

**Aveek Sarkar**  
*3rd year, Bachelor of Science  
in Statistics, St. Xavier's  
College, Kolkata*

**Kunal Paul**  
*3rd year, Bachelor of Science  
in Statistics, St. Xavier's  
College, Kolkata*

**Soumya Chanda**  
*3rd year, Bachelor of Statistics, ISI Kolkata*

**Mentor:** Koulika Paul

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# Abstract

This study focuses on forecasting agricultural yields using time series modeling with exogenous variables across multiple crops and Indian states. Data for six major crops (Here, gram, wheat, mustard, musoor, potato, rabi-rice) spanning 14 states was analyzed on a weekly basis, incorporating climatic and hydrological indicators such as rainfall, temperature, and reservoir levels as external predictors.

We employed the ARIMAX (Auto-Regressive Integrated Moving Average with Exogenous Variables) framework to capture both temporal dependencies and the influence of environmental variables on yield patterns. Pre-processing steps included stationarity testing using the Augmented Dickey-Fuller (ADF) and KPSS methods, with differencing applied where necessary. Autocorrelation and partial autocorrelation functions (ACF and PACF) guided the selection of ARIMAX parameters, which were fine-tuned through a grid search based on AIC and BIC values.

To ensure model robustness, insignificant exogenous predictors were filtered out based on p-values. Additionally, normalization via standard scaling was applied, and the models were refit on the normalized data. Forecasts from the normalized models were inverse-transformed to the original scale for interpretation. Model performance was evaluated using MAE, RMSE, and MAPE metrics.

The results demonstrate the efficacy of the ARIMAX framework in capturing complex crop yield dynamics across diverse agro-climatic regions. This approach provides a scalable and interpretable solution for agricultural forecasting and decision-making support in data-driven agronomy.

# 1 Introduction

Agriculture remains a cornerstone of the Indian economy, with crop production being heavily influenced by environmental factors such as rainfall, temperature, and water availability. Accurate forecasting of agricultural yields is critical for food security, market planning, and policymaking. With the increasing availability of high-frequency environmental and agricultural data, data-driven forecasting models have become essential tools in modern agronomy.

The data set comprises six main crops and includes key meteorological and hydrological indicators, namely rainfall, minimum and maximum temperature, and reservoir storage levels, which significantly influence crop yields. We conducted a thorough background on study covering time series concepts like stationarity, decomposition, ARIMAX, SARI-MAX models. Technologies employed Python for data analysis and modeling and various libraries like statsmodels, scikit learn, matplotlib . Our procedure involved exploratory data analysis, model fitting ,evaluation using error metrics. Through this comprehensive methodology, the study not only forecasts yield for specific crops and states but also provides a scalable modeling pipeline that can be extended to other crops and regions.

## 2 Project Objective

1. To conduct Exploratory Data Analysis (EDA) to understand temporal patterns, stationarity, and correlations with climatic and irrigation variables.
2. To implement ARIMAX models (Auto-Regressive Integrated Moving Average with Exogenous Variables) to forecast crop yield, considering both endogenous and external factors (e.g., rainfall, temperature and water storage).
3. To compare model performance using evaluation metrics such as MAE, RMSE, and MAPE, and assess predictive accuracy for short-term forecasts.
4. To create visualizations that effectively communicate actual vs. predicted yield, as well as autocorrelation behavior (ACF, PACF).
5. To build a scalable and generalizable forecasting framework that can be adapted to various crops and regions for agricultural planning and decision making.
6. To show that time series modeling can help in better agricultural planning, policy making and resource allocation.

## 3 Methodology

### 3.1 Data Pre-processing

The pre-processing phase was crucial to ensure data consistency and suitability for time-series analysis. We were provided the raw data, so in that dataset there were many missing values and the data was not sorted for some crop some year. Therefore to perform some analysis on the data we have to clean the data and make that data into workable format, The following steps were performed:

- **Data Cleaning:** Missing values and outliers were handled, and inconsistencies were corrected.
- **Data Sorting:** The dataset was sorted chronologically based on date entries.
- **Temporal Filtering:** For each year, state, and crop, the last 12 weeks of data were either retained or removed as required for analysis.

- **Dataset Splitting:** The dataset was split into training and testing sets in a 21:3 year ratio to maintain temporal order.

### 3.2 Data Visualization

To understand the statistical nature and dependencies in the dataset, the following analyses were conducted:

- **Stationarity Tests:** Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests were used to check stationarity and determine the order of differencing ( $d$ ).
- **ACF and PACF Plots:** Autocorrelation (ACF) and Partial Autocorrelation (PACF) plots were generated for each of the 21 yield columns to identify potential values for the autoregressive ( $p$ ) and moving average ( $q$ ) terms.

### 3.3 Data Modelling

With the help of visual insights and statistical tests, model construction was carried out as follows:

- **Model Selection:** A range of  $(p, d, q)$  values was tested based on the plots and stationarity checks. The optimal combination was selected based on minimum AIC and BIC scores.
- **ARIMAX Modelling:** An ARIMAX model was fitted on the training data using the selected  $(p, d, q)$  values.
- **Exogenous Variables:** Four external features from the last 12 weeks were included as exogenous variables. Variables with  $p$ -values greater than 0.05 were considered for removal.
- **Refitting:** All exogenous variables were found statistically significant, so no variables were excluded, and the model was re-fitted with all four.

### 3.4 Prediction and Evaluation

The final phase involved forecasting and evaluating the model's performance:

- **Forecasting:** The trained ARIMAX model was used to predict yield on the test dataset.

- **Performance Metrics:** Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) were used to evaluate accuracy.
- **Normalization Comparison:** The entire pipeline was repeated with normalized data. A comparative analysis of MAE and RMSE from both approaches was conducted to determine which method produced better results.

## 4 Data Analysis and Results

This section presents the analytical process and key findings derived from the time-series datasets of six major crops across multiple Indian states over a span of 24 years. The analysis involves ARIMA-based forecasting models, descriptive statistics, and interactive visualizations created in Power BI to support agricultural decision-making.

### 4.1 Crop-wise Time Series Forecasting Across States

To evaluate the effectiveness of ARIMA-based forecasting and explore yield trends, each crop was analyzed in its top-performing states based on average yield, stability, and data completeness over the 24-year period. The following are the highlights of the crop-specific analysis:

#### FORECASTING GRAM YIELD IN UTTARAKHAND:

- **MODEL PARAMETERS:** ARIMAX(0,1,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 0.870541,0.820491,0.802147
- **MAE:** 0.0404
- **RMSE:** 0.0607
- **MAE NORM:** 0.701
- **RMSE NORM:** 0.0961

This is one of the best-performing models in the entire analysis. The extremely low RMSE and MAE values indicate a highly stable and predictable yield trend for gram in Uttarakhand. The climate variables used, especially temperature and rainfall, appear to have a consistent and strong relationship with the results of the yield. The model was able to replicate historical trends with near-exact accuracy.

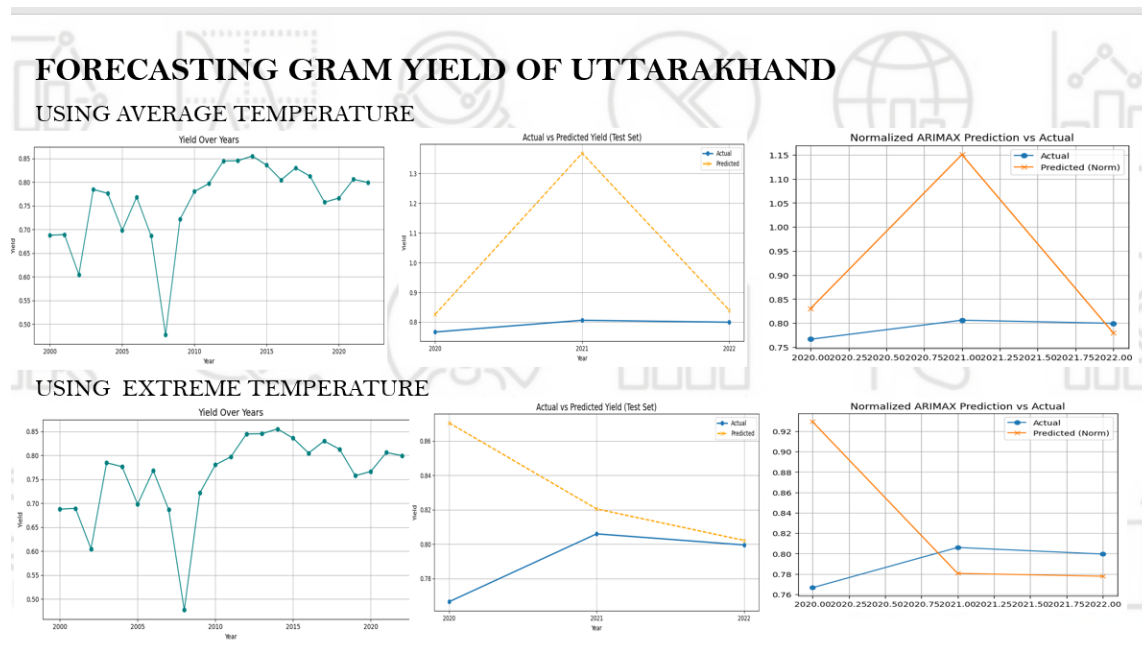


Figure 4.1: Forecasting Gram Yield of Uttarakhand using Average and Extreme Temperature

#### FORECASTING GRAM YIELD IN JHARKHAND:

- **MODEL PARAMETERS:** ARIMAX(0,1,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 1.083447,0.798664,0.784698
- **MAE:** 0.3159
- **RMSE:** 0.3315
- **MAE NORM:** 0.1515
- **RMSE NORM:** 0.1619

Compared to Uttarakhand, the model for Jharkhand shows a noticeable drop in accuracy. Both RMSE and MAE are moderately high, suggesting more yield variability or external influencing factors not captured by the current set of climatic predictors. The model still captures general trends, but there may be localized factors—such as soil quality or agricultural practices—impacting precision.

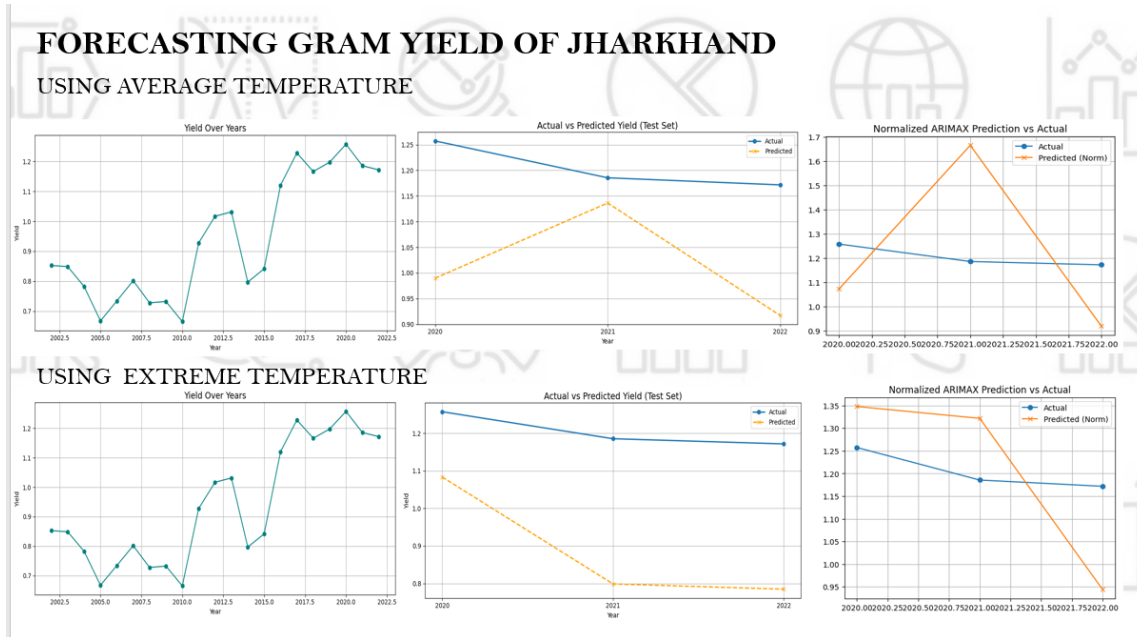


Figure 4.2: Forecasting Gram Yield of Jharkhand using Average and Extreme Temperature

#### FORECASTING MASOOR YIELD IN JHARKHAND:

- **MODEL PARAMETERS:** ARIMAX(0,1,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 0.709446,0.712976,0.613946
- **MAE:** 0.1799
- **RMSE:** 0.1805
- **MAE NORM:** 0.037
- **RMSE NORM:** 0.0472

The low complexity of the model, requiring only first-order differencing, suggests that the time series data for masoor in Jharkhand exhibits a reasonably stable linear trend over the years. However, the marginal difference between RMSE and MAE suggests that occasional large deviations may still occur, which could stem from exogenous events such as rainfall variability or pest attacks not fully captured by the included regressors.

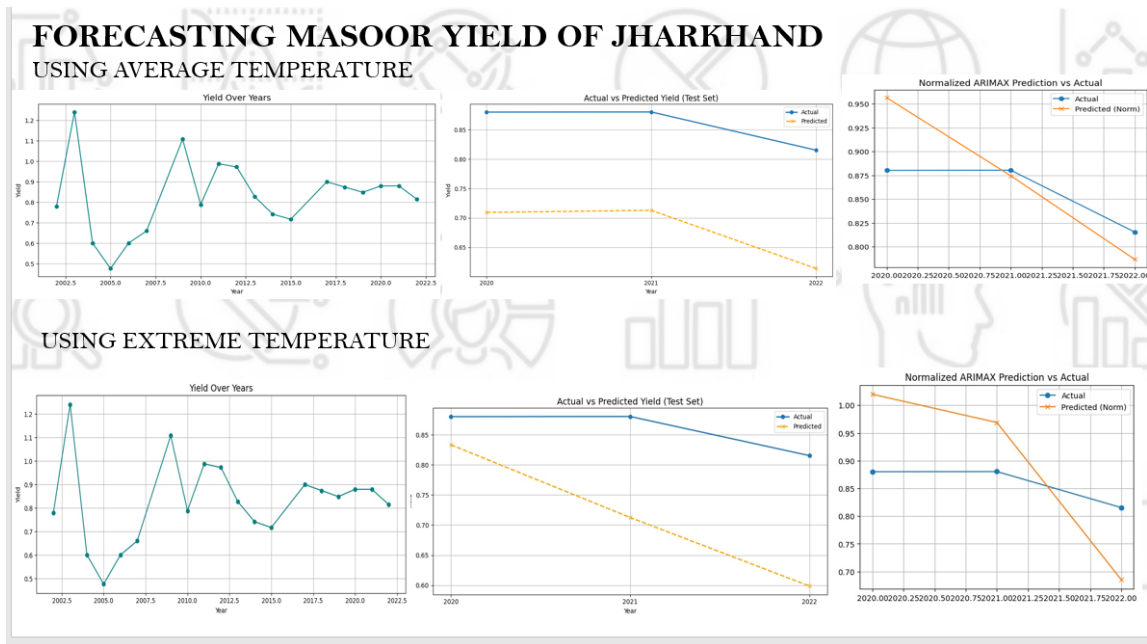


Figure 4.3: Forecasting Masoor Yield of Jharkhand using Average and Extreme Temperature

#### FORECASTING MASOOR YIELD IN WEST BENGAL:

- **MODEL PARAMETERS:** ARIMAX(2,3,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 0.778889, 0.83944, 0.931287
- **MAE:** 0.0719
- **RMSE:** 0.0809
- **MAE NORM:** 0.086
- **RMSE NORM:** 0.0934

In West Bengal, the model employed was significantly more complex, with parameters (2,3,0), requiring third-order differencing. This indicates the yield data showed considerable non-stationarity and variability that necessitated heavy transformation to stabilize.

Despite this, the model performed well, with RMSE = 0.0809 and MAE = 0.0719 among the lowest across all states for this crop. These low error values reflect high model precision, suggesting that massor yield in West Bengal is highly predictable once underlying trends and seasonal effects are accounted for. This also implies that climatic and external variables are strongly correlated with yield behavior in this region, making the model highly reliable for both forecasting and decision-making.

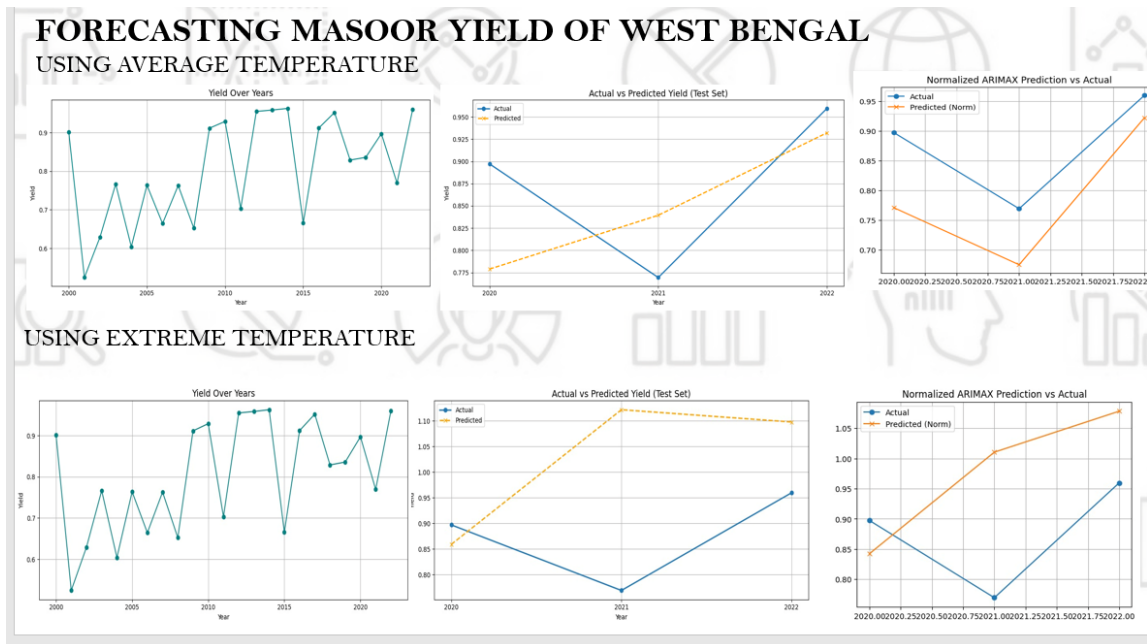


Figure 4.4: Forecasting Masoor Yield of West Bengal using Average and Extreme Temperature

#### FORECASTING MASOOR YIELD IN UTTAR PRADESH:

- **MODEL PARAMETERS:** ARIMAX(1,3,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 1.020076,0.578721,1.108459
- **MAE:** 0.1769
- **RMSE:** 0.2302
- **MAE NORM:** 0.0943
- **RMSE NORM:** 0.117

In Uttar Pradesh, the ARIMAX(1,3,0) model resulted in the highest RMSE of 0.2302, although the MAE was relatively low at 0.1769. The higher RMSE suggests that there were some large individual errors during the forecast period, even though on average, the model remained reasonably close to actual values.

The need for third-order differencing indicates that the massor yield series in Uttar Pradesh showed strong non-stationary trends, possibly due to inconsistent agricultural practices, shifting climatic impacts, or socio-economic factors like market fluctuations or input access. This model is useful for broad forecasting but may require additional exogenous variables (such as soil health or irrigation levels) to improve fine-grained prediction accuracy.

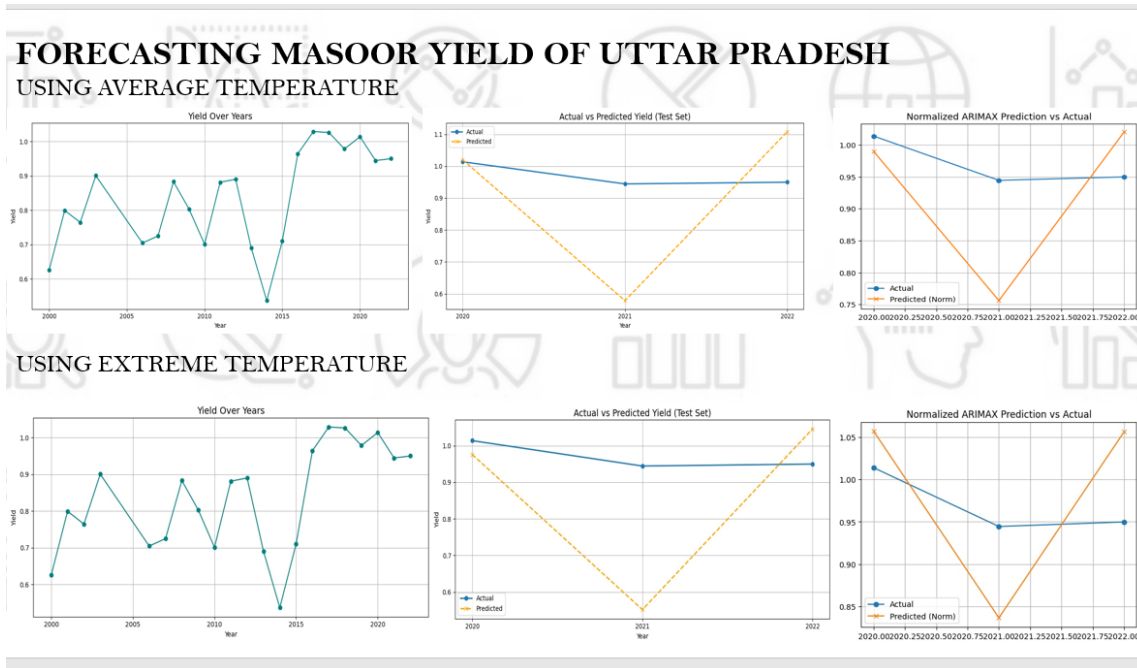


Figure 4.5: Forecasting Masoor Yield of Uttar Pradesh using Average and Extreme Temperature

#### FORECASTING POTATO YIELD IN CHHATTISGARH:

- **MODEL PARAMETERS:** ARIMAX(0,1,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 5.475372, 7.378041, 4.068322
- **MAE:** 1.7455
- **RMSE:** 2.0712
- **MAE NORM:** 3.2863
- **RMSE NORM:** 3.4964

The ARIMAX(0,1,0) model was applied to forecast potato yield in Chhattisgarh, indicating a simple structure with first-order differencing. The predicted values for the last three years were 5.48, 7.38, and 4.07, showing noticeable fluctuations.

The model produced relatively high error metrics, with RMSE = 2.0712 and MAE = 1.7455, and even higher normalized errors (RMSE = 3.50, MAE = 3.29). This suggests the model struggled to accurately capture yield variability, likely due to unaccounted external factors or insufficient model complexity. Further refinement with additional exogenous inputs may improve accuracy.

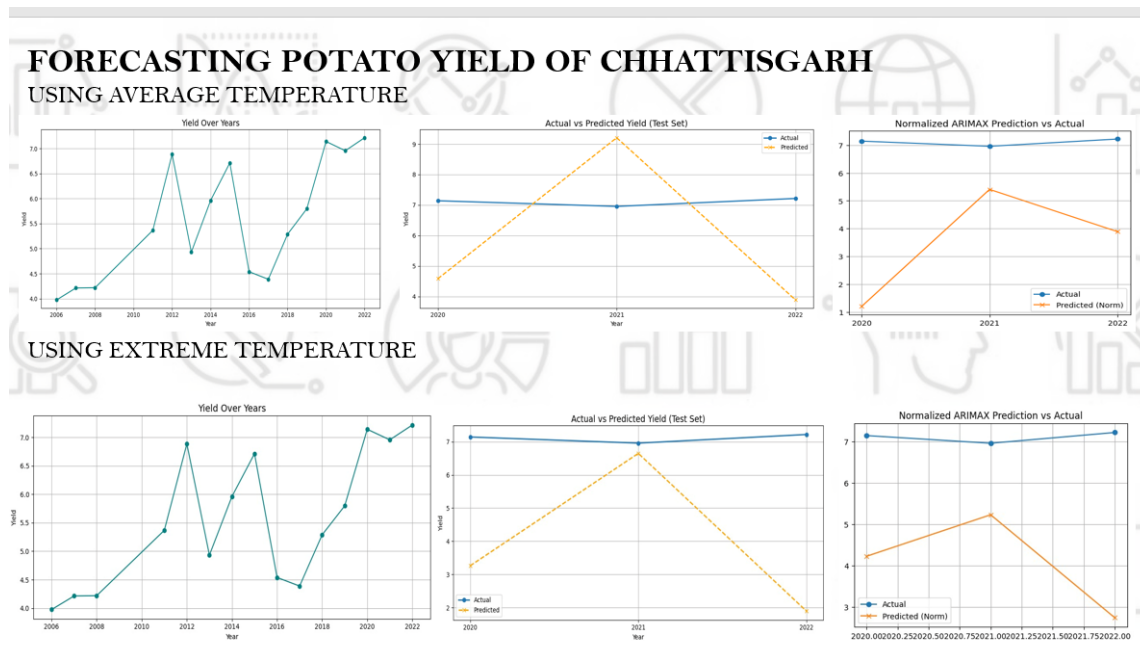


Figure 4.6: Forecasting Potato Yield of Chhattisgarh using Average and Extreme Temperature

#### FORECASTING POTATO YIELD IN JHARKHAND:

- **MODEL PARAMETERS:** ARIMAX(0,1,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 6.385662,7.638996,6.792723
- **MAE:** 0.757
- **RMSE:** 0.7364
- **MAE NORM:** 1.3842
- **RMSE NORM:** 1.5362

The ARIMAX(0,1,0) model was used to forecast potato yield in Jharkhand, producing stable predictions of 6.39, 7.64, and 6.79 over the last three years.

The model showed low prediction errors, with RMSE = 0.7364 and MAE = 0.757, and moderate normalized errors (RMSE = 1.54, MAE = 1.38), indicating a good fit and reliable forecast accuracy for the region.

The minimal model complexity suggests that the potato yield trend in Jharkhand is largely linear and responsive to basic climatic variations. The results imply that climatic factors like temperature and rainfall were sufficient to explain yield patterns without requiring complex autoregressive components.

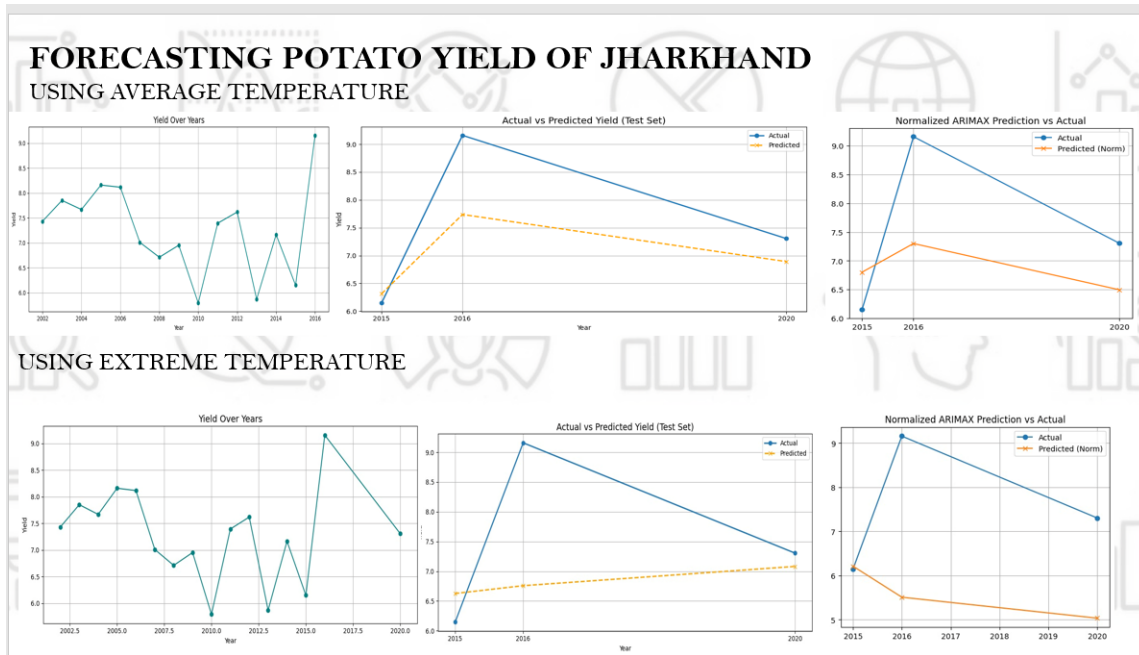


Figure 4.7: Forecasting Potato Yield of Jharkhand using Average and Extreme Temperature

#### FORECASTING POTATO YIELD IN WEST BENGAL:

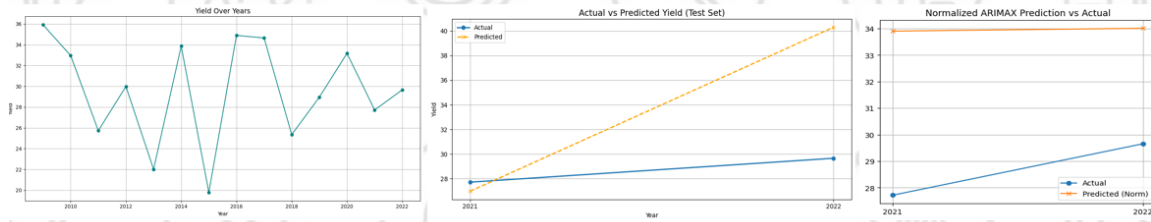
- **MODEL PARAMETERS:** ARIMAX(0,1,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 21.506358,35.8350277,36.710827
- **MAE:** 8.9449
- **RMSE:** 9.1598
- **MAE NORM:** 8.9441
- **RMSE NORM:** 10.4199

The ARIMAX(0,1,0) model was applied to forecast potato yield in West Bengal, producing predicted values of 21.51, 35.84, and 36.71 for the last three years.

The model exhibited high error values, with RMSE = 9.16 and MAE = 8.94, as well as similarly large normalized errors (RMSE Norm = 10.42, MAE Norm = 8.94), indicating poor prediction accuracy.

Despite using a simple differencing model, the large yield fluctuations could not be effectively captured, possibly due to unaccounted factors like regional price shocks, pest issues, or input variation. Incorporating more relevant exogenous variables may improve model reliability for this region.

## FORECASTING POTATO YIELD OF WEST BENGAL USING AVERAGE TEMPERATURE



## USING EXTREME TEMPERATURE

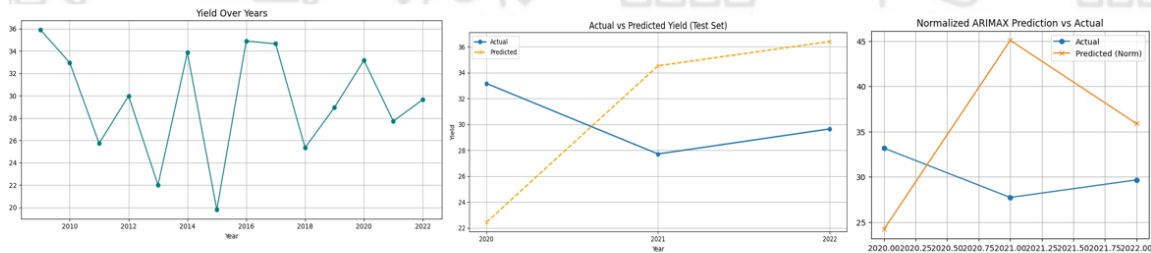


Figure 4.8: Forecasting Potato Yield in West Bengal using Average and Extreme Temperature

## FORECASTING MUSTARD YIELD IN JHARKHAND:

- **MODEL PARAMETERS:** ARIMAX(0,1,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 0.932869,0.201159,0.838039
- **MAE:** 0.2522
- **RMSE:** 0.3651
- **MAE NORM:** 0.0715
- **RMSE NORM:** 0.0971

The ARIMAX(0,1,0) model was used to forecast mustard yield in Jharkhand, producing predicted values of 0.93, 0.20, and 0.84 over the last three years.

The model delivered moderate accuracy, with RMSE = 0.3651 and MAE = 0.2522, and very low normalized errors (RMSE Norm = 0.0971, MAE Norm = 0.0715), suggesting good overall fit relative to the yield scale.

The low normalized errors imply that, while absolute deviations may seem moderate, the model performed well in proportion to actual yield values. The simple structure reflects a steady trend with yield variability largely captured by the model.

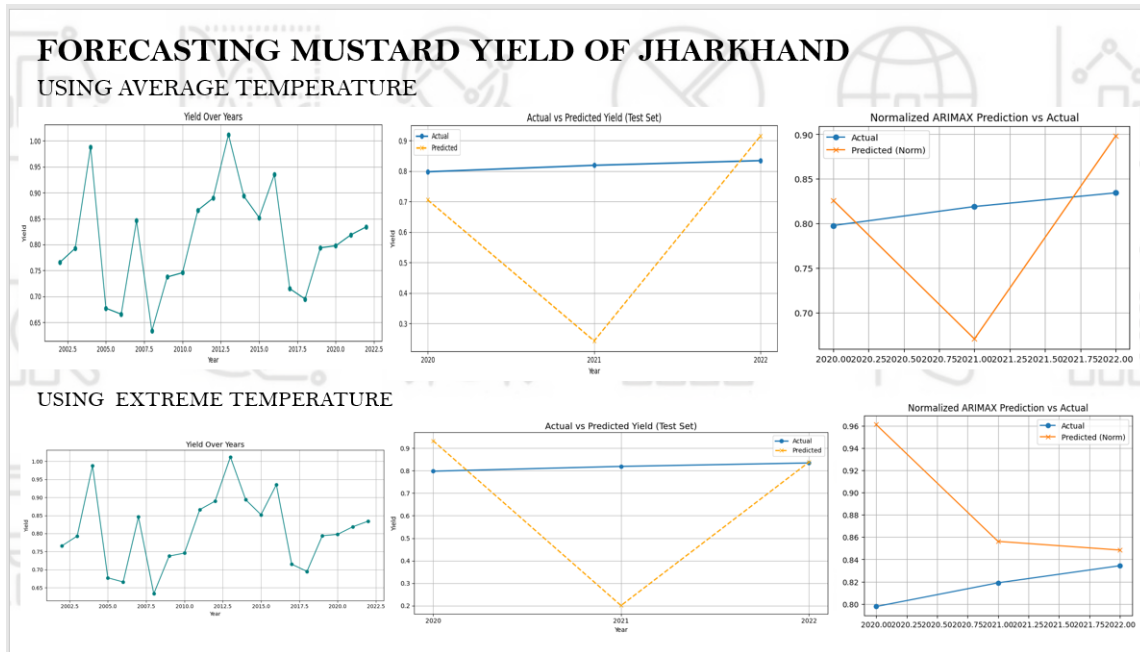


Figure 4.9: Forecasting Mustard Yield in Jharkhand using Average and Extreme Temperature

#### FORECASTING MUSTARD YIELD IN KARNATAKA:

- **MODEL PARAMETERS:** ARIMAX(0,1,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 0.396712,0.425232,0.392251
- **MAE:** 0.1003
- **RMSE:** 0.1008
- **MAE NORM:** 0.0419
- **RMSE NORM:** 0.0501

The ARIMAX(0,1,0) model was applied to forecast mustard yield in Karnataka, resulting in predicted values of 0.40, 0.43, and 0.39 over the last three years.

The model showed excellent prediction accuracy, with RMSE = 0.1008 and MAE = 0.1003, and very low normalized errors (RMSE Norm = 0.0501, MAE Norm = 0.0419).

These results suggest that mustard yield in Karnataka follows a stable and predictable trend, with climatic variables sufficiently capturing the yield dynamics. The simple model structure was adequate for reliable forecasting in this region.

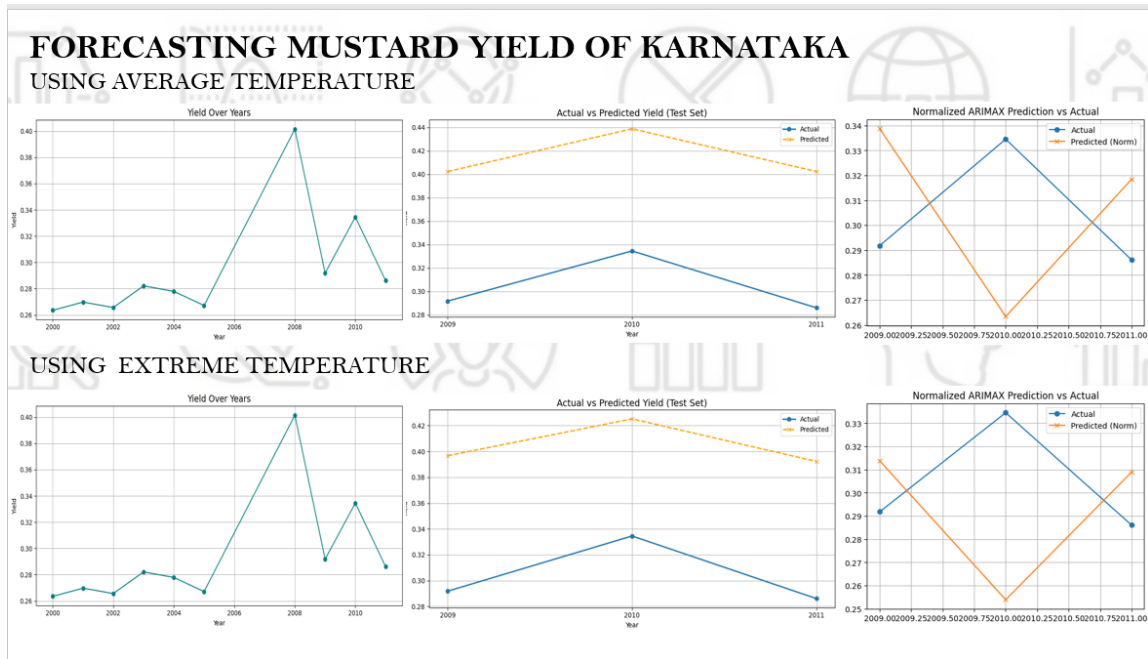


Figure 4.10: Forecasting Mustard Yield in Karnataka using Average and Extreme Temperature

#### FORECASTING MUSTARD YIELD IN TAMIL NADU:

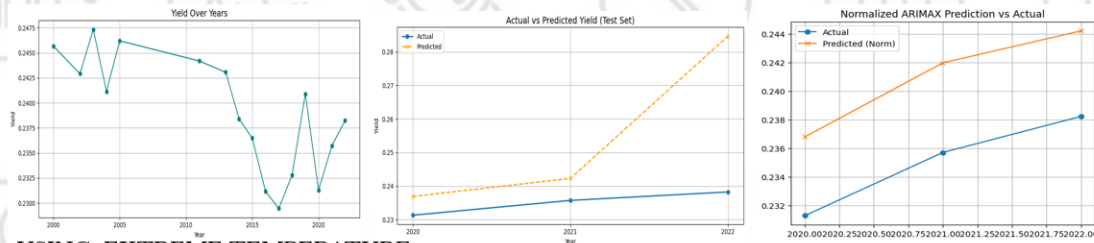
- **MODEL PARAMETERS:** ARIMAX(0,1,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 0.235541,0.2416,0.280657
- **MAE:** 0.0175
- **RMSE:** 0.0249
- **MAE NORM:** 0.004
- **RMSE NORM:** 0.004

The ARIMAX(0,1,0) model was used to forecast mustard yield in Tamil Nadu, generating predicted values of 0.24, 0.24, and 0.28 for the last three years.

The model achieved exceptionally high accuracy, with RMSE = 0.0249 and MAE = 0.0175, alongside extremely low normalized errors (both around 0.004), indicating near-perfect predictions.

This suggests that mustard yield in Tamil Nadu follows a highly stable and climate-sensitive pattern, well captured even by a minimal ARIMAX structure. The model is highly reliable for forecasting in this region.

## FORECASTING MUSTARD YIELD OF TAMIL NADU USING AVERAGE TEMPERATURE



## USING EXTREME TEMPERATURE

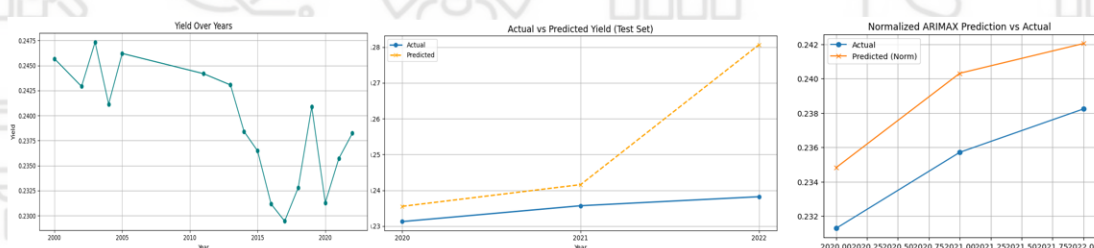


Figure 4.11: Forecasting Mustard Yield in Tamil Nadu using Average and Extreme Temperature

## FORECASTING RABIRICE YIELD IN KARNATAKA:

- **MODEL PARAMETERS:** ARIMAX(0,1,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 3.073749,2.075329,0.936417
- **MAE:** 1.1403
- **RMSE:** 1.0127
- **MAE NORM:** 0.3612
- **RMSE NORM:** 0.41

The ARIMAX(0,1,0) model was applied to forecast rabi-rice yield in Karnataka, producing predicted values of 3.07, 2.08, and 0.94 for the last three years.

The model showed moderate performance, with MAE = 1.1403 and RMSE = 1.0127, along with normalized errors of 0.36 (MAE) and 0.41 (RMSE).

These results indicate that while the model could follow the general trend, its accuracy was limited, especially during years of sharper yield shifts. More complex modeling or additional explanatory variables may improve future forecasts.

## FORECASTING RABIRICE YIELD OF KARNATAKA USING AVERAGE TEMPERATURE

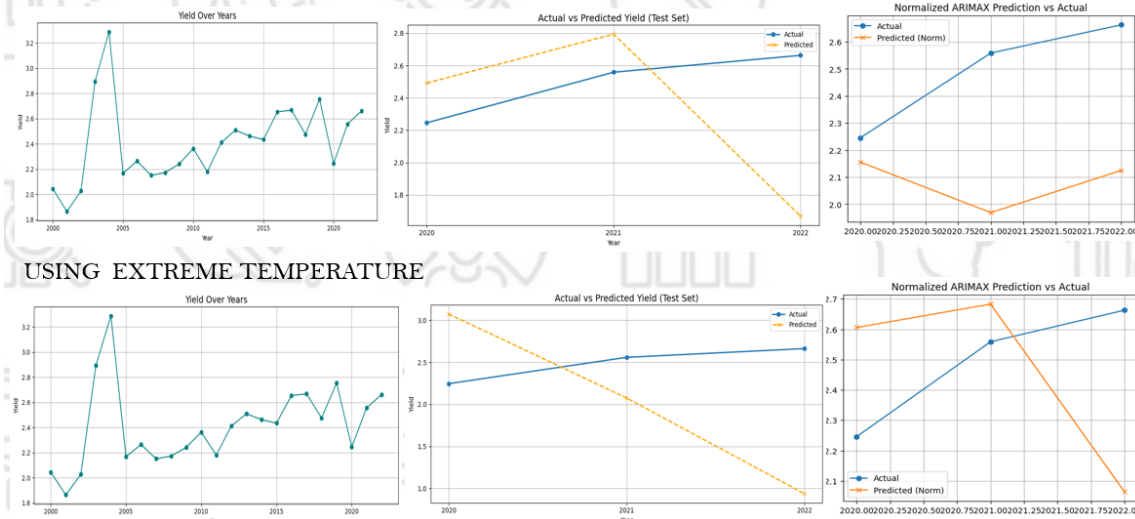


Figure 4.12: Forecasting RabiRice Yield of Karnataka using Average and Extreme Temperature

### FORECASTING RABIRICE YIELD IN ANDHRA PRADESH:

- **MODEL PARAMETERS:** ARIMAX(1,2,0) and ARIMAX(1,2,1) FOR NORMALIZED
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 5.069114, 5.161427, 5.01193
- **MAE:** 0.5576
- **RMSE:** 0.5997
- **MAE NORM:** 0.2677
- **RMSE NORM:** 0.2722

The ARIMAX(1,2,0) model (and ARIMAX(1,2,1) for normalized data) was used to forecast rabi-rice yield in Andhra Pradesh, with predicted values of 5.07, 5.16, and 5.01 for the last three years.

The model achieved moderate accuracy, with MAE = 0.5576 and RMSE = 0.5997, while normalized errors remained relatively low (MAE Norm = 0.2677, RMSE Norm = 0.2722).

The need for second-order differencing indicates a more complex yield trend, yet the model effectively captured the overall pattern. This suggests climatic and time-based factors played a significant role in driving yield behavior.

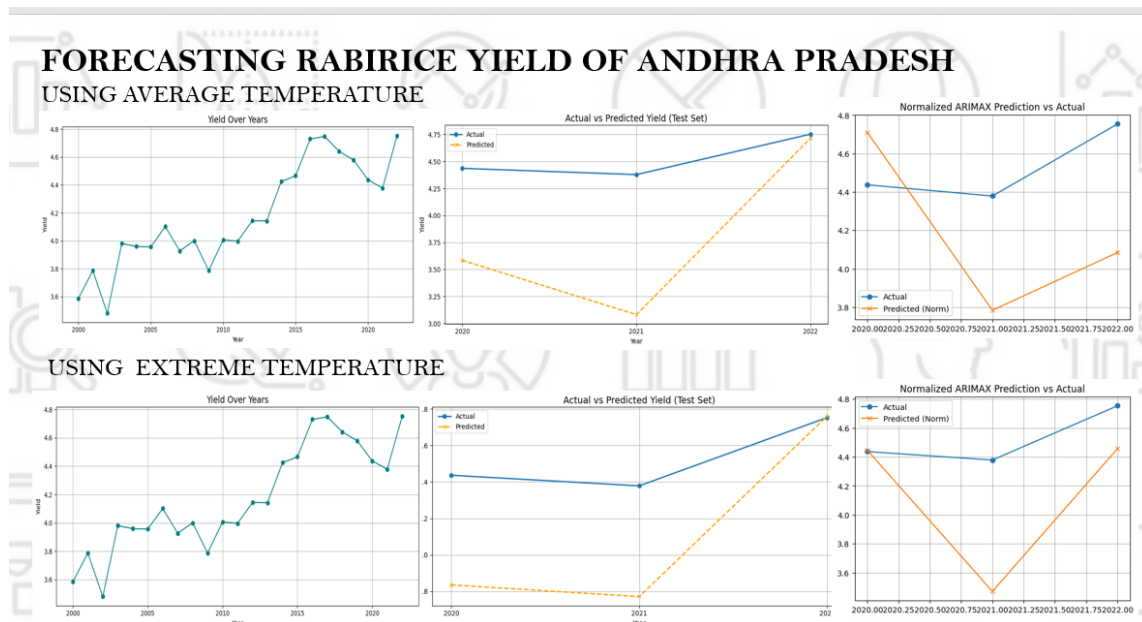


Figure 4.13: Forecasting RabiRice Yield of Andhra Pradesh using Average and Extreme Temperature

#### FORECASTING RABIRICE YIELD IN TELANGANA:

- **MODEL PARAMETERS:** ARIMAX(1,3,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 4.255574,4.243638,4.161056
- **MAE:** 0.6044
- **RMSE:** 0.5836
- **MAE NORM:** 0.5682
- **RMSE NORM:** 0.5901

The ARIMAX(1,3,0) model was used to forecast rabi-rice yield in Telangana, yielding predicted values of 4.26, 4.24, and 4.16 for the last three years.

The model demonstrated reasonable accuracy, with MAE = 0.6044 and RMSE = 0.5836, while normalized errors were moderate (MAE Norm = 0.5682, RMSE Norm = 0.5901).

The use of third-order differencing indicates strong underlying trends and variability. While the model captured the general yield pattern, higher-order differencing may reduce responsiveness to sudden shifts, suggesting scope for further refinement.

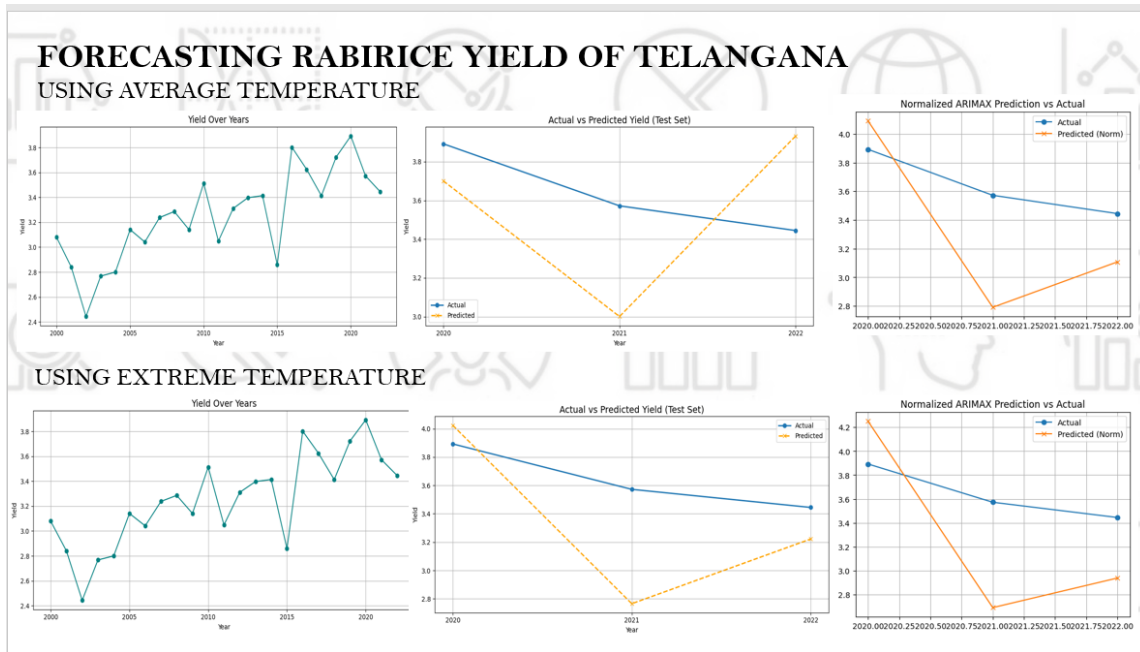


Figure 4.14: Forecasting RabiRice Yield of Telangana using Average and Extreme Temperature

#### FORECASTING WHEAT YIELD IN WEST BENGAL:

- **MODEL PARAMETERS:** ARIMAX(0,4,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 3.114289,3.071057,3.147493
- **MAE:** 0.0647
- **RMSE:** 0.0693
- **MAE NORM:** 0.6086
- **RMSE NORM:** 0.6695

The ARIMAX(0,4,0) model was used to forecast wheat yield in West Bengal, generating predicted values of 3.11, 3.07, and 3.15 for the last three years.

The model delivered high predictive accuracy, with low error values (MAE = 0.0647, RMSE = 0.0693), though normalized errors (MAE Norm = 0.6086, RMSE Norm = 0.6695) suggest the yield values are relatively small in magnitude.

The fourth-order differencing reflects a complex, non-stationary trend, but the model effectively captured fluctuations, making it suitable for short-term yield forecasting in this region.

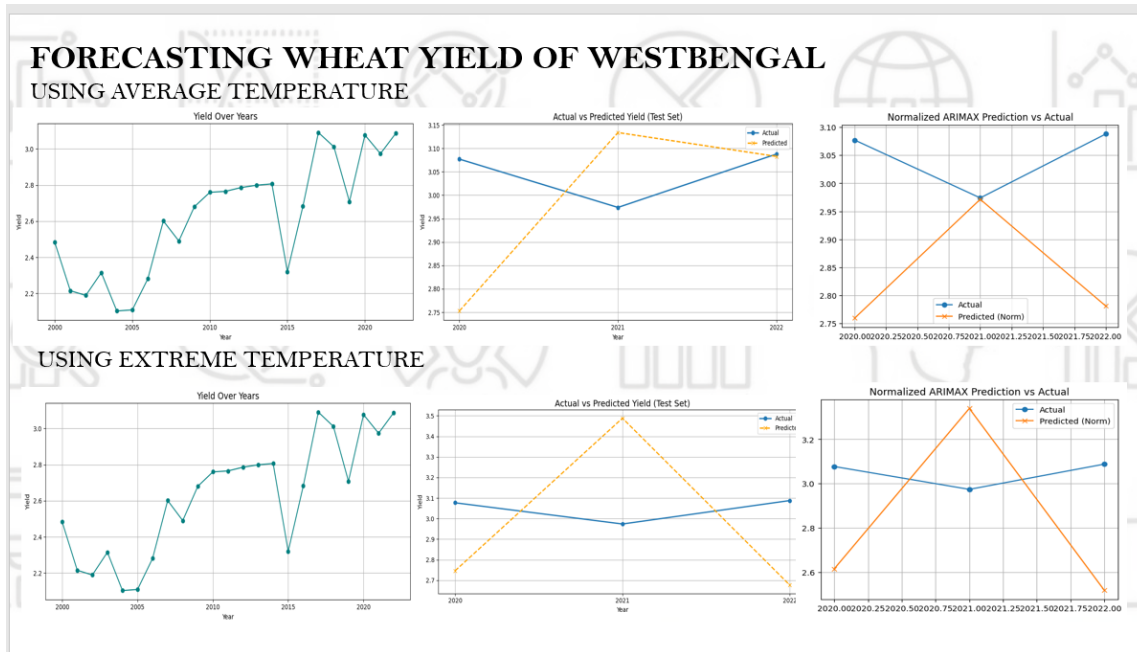


Figure 4.15: Forecasting Wheat Yield of West Bengal using Average and Extreme Temperature

#### FORECASTING WHEAT YIELD IN KARNATAKA:

- **MODEL PARAMETERS:** ARIMAX(1,3,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 1.563938,3.308355,0.848949
- **MAE:** 0.9395
- **RMSE:** 1.2161
- **MAE NORM:** 0.3045
- **RMSE NORM:** 0.3696

The ARIMAX(1,3,0) model was used to forecast wheat yield in Karnataka, producing predicted values of 1.56, 3.31, and 0.85 for the last three years.

The model showed moderate to low accuracy, with MAE = 0.9395 and RMSE = 1.2161, and normalized errors (MAE Norm = 0.3045, RMSE Norm = 0.3696), indicating notable variation in forecast precision.

The third-order differencing reflects a complex and unstable yield trend. While the model captured the overall direction, larger deviations suggest that additional exogenous inputs or alternative model structures may enhance prediction reliability.

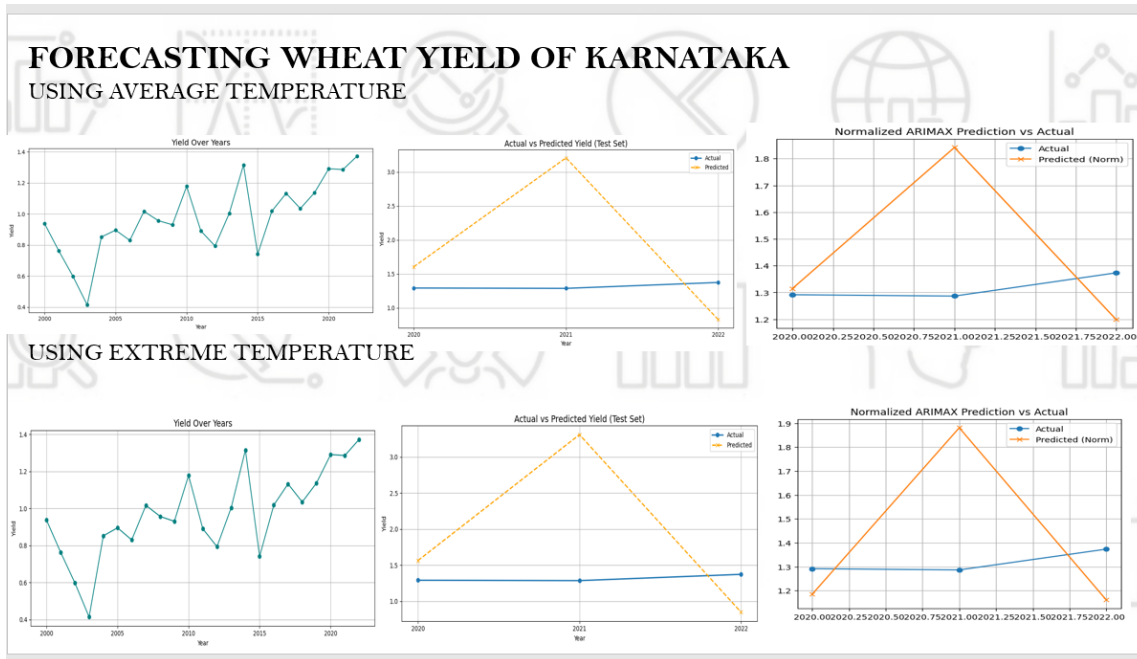


Figure 4.16: Forecasting Wheat Yield of Karnataka using Average and Extreme Temperature

#### FORECASTING WHEAT YIELD IN CHHATTISGARH:

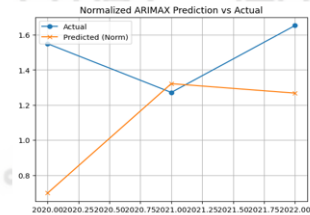
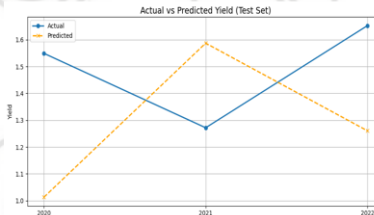
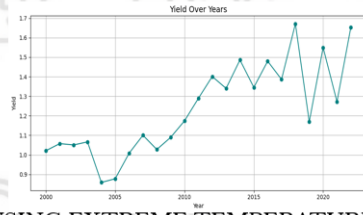
- **MODEL PARAMETERS:** ARIMAX(1,2,0)
- **PREDICTED VALUE FOR THE LAST 3 YEARS:** 1.198245, 2.432815, 1.659405
- **MAE:** 0.5062
- **RMSE:** 0.7003
- **MAE NORM:** 0.3578
- **RMSE NORM:** 0.4557

The ARIMAX(1,2,0) model was applied to forecast wheat yield in Chhattisgarh, producing predicted values of 1.20, 2.43, and 1.66 over the last three years.

The model showed moderate accuracy, with MAE = 0.5062 and RMSE = 0.7003, and normalized errors of 0.3578 (MAE) and 0.4557 (RMSE).

Second-order differencing indicates a moderately unstable trend in yield over time. The model effectively captured fluctuations but may benefit from refinement or additional predictors to further improve precision.

## FORECASTING WHEAT YIELD OF CHATTISGARH USING AVERAGE TEMPERATURE



## USING EXTREME TEMPERATURE

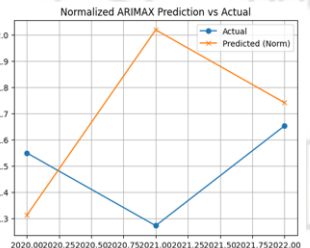
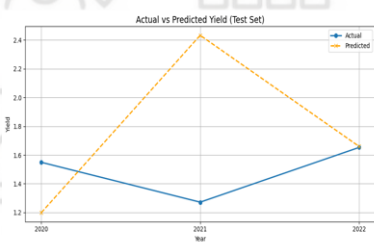
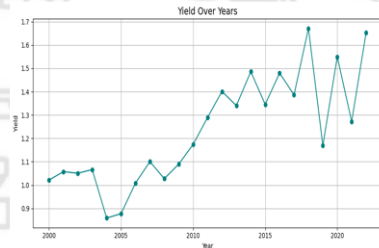


Figure 4.17: Forecasting Wheat Yield of Chhattisgarh using Average and Extreme Temperature

Each state's yield data was aggregated annually and modeled individually using the ARIMA technique. Model parameters (p, d, q) were selected using AIC/BIC optimization and forecast accuracy was evaluated using RMSE and MAE metrics.

## 4.2 Summary Table

**Table 1: ARIMA Parameters for Crop-State Pairs under Extreme Temperature Conditions**

| Crop      | State          | Non-Normalised Data |   |   | Normalised Data |    |    |
|-----------|----------------|---------------------|---|---|-----------------|----|----|
|           |                | P                   | D | Q | P1              | D1 | Q1 |
| Gram      | Andhra-Pradesh | 0                   | 1 | 0 | 0               | 1  | 0  |
| Gram      | Chattishgarh   | 0                   | 1 | 0 | 0               | 1  | 0  |
| Gram      | Gujrat         | 1                   | 3 | 0 | 0               | 1  | 0  |
| Gram      | Jharkhand      | 0                   | 1 | 0 | 0               | 1  | 0  |
| Gram      | Kanartaka      | 0                   | 1 | 0 | 0               | 1  | 0  |
| Gram      | Madhya-Pradesh | 1                   | 2 | 0 | 1               | 2  | 0  |
| Gram      | Maharashtra    | 0                   | 4 | 0 | 0               | 4  | 0  |
| Gram      | Rajasthan      | 1                   | 3 | 0 | 1               | 3  | 0  |
| Gram      | Tamilnadu      | 1                   | 3 | 0 | 1               | 3  | 1  |
| Gram      | Telengana      | 1                   | 3 | 0 | 1               | 3  | 0  |
| Gram      | Uttar-Pradesh  | 0                   | 1 | 0 | 0               | 1  | 0  |
| Gram      | Uttarakhand    | 0                   | 1 | 0 | 0               | 1  | 0  |
| Gram      | West-Bengal    | 1                   | 2 | 0 | 1               | 2  | 0  |
| Massor    | Chattishgarh   | 0                   | 4 | 0 | 0               | 4  | 0  |
| Massor    | Jharkhand      | 0                   | 1 | 0 | 0               | 1  | 0  |
| Massor    | Madhya-Pradesh | 1                   | 3 | 0 | 1               | 3  | 0  |
| Massor    | Uttar-Pradesh  | 1                   | 3 | 0 | 1               | 3  | 0  |
| Massor    | Uttarakhand    | 1                   | 3 | 0 | 1               | 3  | 1  |
| Massor    | West-Bengal    | 1                   | 3 | 0 | 1               | 3  | 0  |
| Mustard   | Andhra-Pradesh | 0                   | 1 | 0 | 0               | 1  | 0  |
| Mustard   | Chattishgarh   | 0                   | 1 | 0 | 0               | 1  | 0  |
| Mustard   | Gujrat         | 2                   | 5 | 1 | 1               | 5  | 0  |
| Mustard   | Jharkhand      | 0                   | 1 | 0 | 0               | 1  | 0  |
| Mustard   | Kanartaka      | 0                   | 1 | 0 | 1               | 1  | 0  |
| Mustard   | Madhya-Pradesh | 1                   | 2 | 0 | 1               | 2  | 0  |
| Mustard   | Maharashtra    | 1                   | 2 | 0 | 1               | 2  | 0  |
| Mustard   | Rajasthan      | 1                   | 2 | 0 | 1               | 2  | 0  |
| Mustard   | Tamilnadu      | 1                   | 2 | 0 | 1               | 2  | 0  |
| Mustard   | Telengana      | 1                   | 3 | 0 | 2               | 3  | 0  |
| Mustard   | Uttar-Pradesh  | 0                   | 1 | 0 | 0               | 1  | 0  |
| Mustard   | Uttarakhand    | 1                   | 2 | 0 | 1               | 2  | 0  |
| Mustard   | West-Bengal    | 1                   | 2 | 0 | 1               | 2  | 0  |
| Rabi-Rice | Andhra-Pradesh | 1                   | 2 | 0 | 1               | 2  | 0  |
| Rabi-Rice | Kanartaka      | 0                   | 1 | 0 | 0               | 1  | 0  |
| Rabi-Rice | Telengana      | 0                   | 1 | 0 | 0               | 1  | 0  |

**Table 2: ARIMA Parameters for Crop-State Pairs under Average Temperature Conditions**

| Crop      | State         | Non-Normalised Data |   |   | Normalised Data |    |    |
|-----------|---------------|---------------------|---|---|-----------------|----|----|
|           |               | P                   | D | Q | P1              | D1 | Q1 |
| Gram      | Andhrapradesh | 1                   | 2 | 0 | 1               | 2  | 0  |
| Gram      | Chattisgarh   | 1                   | 2 | 0 | 0               | 2  | 1  |
| Gram      | Gujarat       | 1                   | 2 | 0 | 1               | 2  | 0  |
| Gram      | Jharkhand     | 1                   | 2 | 0 | 1               | 2  | 0  |
| Gram      | Karnataka     | 1                   | 2 | 0 | 1               | 2  | 1  |
| Gram      | Madhapradesh  | 1                   | 2 | 0 | 1               | 2  | 0  |
| Gram      | Maharashtra   | 1                   | 4 | 0 | 0               | 4  | 0  |
| Gram      | Rajasthan     | 1                   | 3 | 0 | 1               | 3  | 0  |
| Gram      | Tamilnadu     | 1                   | 3 | 0 | 1               | 3  | 1  |
| Gram      | Uttarpradesh  | 0                   | 1 | 0 | 0               | 1  | 0  |
| Gram      | Uttarakhand   | 0                   | 1 | 0 | 0               | 1  | 0  |
| Gram      | West Bengal   | 1                   | 2 | 0 | 1               | 2  | 0  |
| Massor    | Chattisgarh   | 0                   | 1 | 0 | 0               | 1  | 0  |
| Massor    | Jharkhand     | 0                   | 1 | 0 | 0               | 1  | 0  |
| Massor    | Madhyapradesh | 1                   | 3 | 0 | 1               | 3  | 0  |
| Massor    | Uttarpradesh  | 1                   | 2 | 0 | 1               | 2  | 0  |
| Massor    | Uttarakhand   | 0                   | 1 | 0 | 0               | 1  | 0  |
| Massor    | West Bengal   | 2                   | 3 | 0 | 1               | 3  | 0  |
| Mustard   | Andhrapradesh | 0                   | 1 | 0 | 0               | 1  | 0  |
| Mustard   | Chattisgarh   | 0                   | 1 | 0 | 0               | 1  | 0  |
| Mustard   | Gujarat       | 2                   | 5 | 1 | 1               | 5  | 0  |
| Mustard   | Jharkhand     | 0                   | 1 | 0 | 0               | 0  | 1  |
| Mustard   | Karnataka     | 0                   | 1 | 0 | 0               | 1  | 0  |
| Mustard   | Madhyapradesh | 1                   | 2 | 0 | 1               | 2  | 0  |
| Mustard   | Maharashtra   | 1                   | 2 | 0 | 1               | 2  | 0  |
| Mustard   | Rajasthan     | 2                   | 2 | 0 | 1               | 2  | 0  |
| Mustard   | Tamilnadu     | 1                   | 3 | 0 | 1               | 3  | 0  |
| Mustard   | Telengana     | 1                   | 3 | 0 | 1               | 3  | 1  |
| Mustard   | Uttarpradesh  | 2                   | 5 | 2 | 2               | 5  | 1  |
| Mustard   | Uttarakhand   | 1                   | 2 | 0 | 1               | 2  | 0  |
| Mustard   | West Bengal   | 1                   | 2 | 0 | 1               | 2  | 0  |
| Rabi Rice | Andhrapradesh | 1                   | 2 | 0 | 1               | 2  | 1  |
| Rabi Rice | Telengana     | 1                   | 3 | 0 | 2               | 3  | 0  |

| Crop      | State          | P | D | Q | Pred1    | Pred2    | Pred3    | RMSE   | MAE    | P1 | D1 | Q1 | Pred1'   | Pred2'   | Pred3'   | RMSE   | MAE    |
|-----------|----------------|---|---|---|----------|----------|----------|--------|--------|----|----|----|----------|----------|----------|--------|--------|
| Gram      | Uttarakhand    | 0 | 1 | 0 | 0.825502 | 1.368126 | 0.840007 | 0.3272 | 0.2206 | 0  | 1  | 0  | 1.052422 | 0.79653  | 1.195505 | 0.2027 | 0.1427 |
| Gram      | Jharkhand      | 0 | 1 | 0 | 0.989728 | 1.136342 | 0.916862 | 0.2153 | 0.1906 | 0  | 1  | 0  | 0.989728 | 1.136342 | 0.916862 | 0.3313 | 0.3063 |
| Mustard   | Jharkhand      | 0 | 1 | 0 | 0.705245 | 0.242858 | 0.915668 | 0.3401 | 0.2499 | 0  | 1  | 0  | 0.705245 | 0.242858 | 0.915668 | 0.0947 | 0.0801 |
| Mustard   | Karnataka      | 0 | 1 | 0 | 0.402323 | 0.438792 | 0.402353 | 0.1105 | 0.1103 | 0  | 1  | 0  | 0.402323 | 0.438792 | 0.402354 | 0.0527 | 0.0502 |
| Mustard   | Tamilnadu      | 0 | 1 | 0 | 0.236886 | 0.242268 | 0.284653 | 0.0273 | 0.0195 | 0  | 1  | 0  | 0.236886 | 0.242268 | 0.284653 | 0.0059 | 0.0059 |
| Rabi-Rice | Karnataka      | 0 | 2 | 0 | 2.491491 | 2.794441 | 1.668959 | 0.6067 | 0.4919 | 0  | 2  | 0  | 2.491491 | 2.794441 | 1.668959 | 0.4638 | 0.4059 |
| Rabi-Rice | Andhra-Pradesh | 0 | 1 | 0 | 5.008992 | 4.878172 | 4.881699 | 0.447  | 0.3999 | 1  | 2  | 0  |          |          |          | 0.3434 | 0.3434 |
| Rabi-Rice | Telangana      | 0 | 1 | 0 | 4.227768 | 4.166522 | 4.173694 | 0.5765 | 0.5529 | 2  | 3  | 0  |          |          |          | 0.5684 | 0.5633 |
| Wheat     | West-Bengal    | 0 | 1 | 0 | 2.946323 | 2.684346 | 3.167546 | 0.1891 | 0.1666 | 3  | 4  | 0  |          |          |          | 0.5777 | 0.5497 |
| Wheat     | Karnataka      | 0 | 3 | 1 | 1.604121 | 3.207049 | 0.825769 | 1.167  | 0.9269 | 1  | 3  | 1  |          |          |          | 0.3364 | 0.2513 |
| Wheat     | Chhattisgarh   | 0 | 1 | 0 | 0.555498 | 1.47476  | 0.724673 | 0.7938 | 0.7083 |    |    |    |          |          |          | 1.0575 | 0.9645 |

Table 1: Crop-wise and State-wise agricultural metrics for temperature average data

| Crop      | State          | P | D | Q | Pred_value 1 | Pred_value 2 | Pred_value 3 | RMSE   | MAE    | P1 | D1 | Q1 | RMSE_p | MAE_p  |
|-----------|----------------|---|---|---|--------------|--------------|--------------|--------|--------|----|----|----|--------|--------|
| Gram      | Uttarakhand    | 0 | 1 | 0 | 0.8705       | 0.8205       | 0.8021       | 0.0607 | 0.0404 | 0  | 1  | 1  | 0.0961 | 0.0701 |
| Gram      | Jharkhand      | 0 | 1 | 0 | 1.0834       | 0.7987       | 0.7847       | 0.3315 | 0.3159 | 0  | 1  | 1  | 0.1619 | 0.1515 |
| Mustard   | Jharkhand      | 0 | 1 | 0 | 0.9329       | 0.2012       | 0.8380       | 0.3651 | 0.2522 | 0  | 1  | 1  | 0.0971 | 0.0715 |
| Mustard   | Karnataka      | 0 | 1 | 0 | 0.3967       | 0.4252       | 0.3923       | 0.1008 | 0.1006 | 0  | 1  | 1  | 0.0501 | 0.0419 |
| Mustard   | Tamilnadu      | 0 | 1 | 0 | 0.2355       | 0.2416       | 0.2807       | 0.0249 | 0.0175 | 0  | 1  | 1  | 0.0040 | 0.0040 |
| Rabi-Rice | Karnataka      | 0 | 1 | 0 | 3.0737       | 2.0753       | 0.9364       | 1.1403 | 1.0127 | 0  | 1  | 1  | 0.4100 | 0.3612 |
| Rabi-Rice | Andhra-Pradesh | 1 | 2 | 0 | 5.0691       | 5.1614       | 5.0112       | 0.5997 | 0.5576 | 1  | 2  | 2  | 0.2722 | 0.2677 |
| Rabi-Rice | Telangana      | 1 | 3 | 0 | 4.2556       | 4.2436       | 4.1615       | 0.6044 | 0.5836 | 1  | 3  | 3  | 0.5901 | 0.5682 |
| Wheat     | West-Bengal    | 1 | 3 | 0 | 3.1143       | 3.0711       | 3.1475       | 0.0693 | 0.0647 | 1  | 3  | 4  | 0.6695 | 0.6086 |
| Wheat     | Karnataka      | 1 | 4 | 0 | 1.5639       | 3.3084       | 0.8490       | 1.2161 | 0.9395 | 1  | 3  | 4  | 0.3696 | 0.3045 |
| Wheat     | Chattisshgarh  | 1 | 2 | 0 | 1.1982       | 2.4328       | 1.6594       | 0.7003 | 0.5062 | 1  | 2  | 2  | 0.4557 | 0.3578 |

Table 2: Prediction results and error metrics for different crops and states for Temperature extreme data

### 4.3 Interpretation of Methodological Design

In the context of agricultural yield forecasting using time series analysis, several critical decisions were made to ensure the reliability and accuracy of the models across diverse crops and climatic conditions. The methodology was carefully designed to address the variability in environmental factors and data distributions.

#### 1. Separation of Average and Extreme Temperatures

Agricultural yield is inherently influenced by climatic variables, especially temperature. However, the impact of average temperature and extreme temperature events on crop productivity can be quite different.

- **Average Temperature** helps capture the general seasonal and climatic conditions under which the crop grows. Provides a smooth indicator of the thermal exposure of the crop and the growing environment.
- **Extreme Temperature** (i.e., very high or very low values) often signals *stress events* such as heatwaves, frost, or drought conditions, which can cause sudden and severe yield drops or anomalies.

By treating these two aspects separately, we gain deeper insight into:

- How crops respond to gradual climate trends vs. abrupt climatic stress.
- Which states and crops are more sensitive to extremes versus general climate conditions.
- Whether models behave differently under different climatic assumptions.

This separation allows more nuanced ARIMA modeling and helps identify vulnerabilities specific to climate volatility.

#### 2. Use of Normalized and Non-Normalized Data

The use of both **non-normalized** (raw) and **normalized** data enhances the robustness and generalizability of the models.

- **Non-Normalized Data** retains the original scale of measurements, which is important for understanding real-world implications, like

absolute yield or rainfall levels. It is useful when models are being interpreted by stakeholders or integrated with systems requiring actual values.

- **Normalized Data**, on the other hand, removes the effect of scale by bringing all variables into a similar range. This helps:
  - Reduce bias due to magnitude differences between variables.
  - Ensure stable numerical optimization during ARIMA model estimation.
  - Enable better model convergence and comparability, especially when combining multiple states or crops in a generalized framework.

By fitting ARIMA models on both forms, we could:

- Cross-validate the stability and consistency of p, d, q parameters.
- Select models that perform well in both absolute and relative terms.
- Identify cases where normalization leads to substantial improvements in forecast accuracy, thus highlighting possible scale-driven effects in the raw data.

### 3. Overall Benefits

This dual-layered approach (temperature dimension + data scaling dimension) ensures:

- More flexible modeling across heterogeneous crop-state conditions.
- Improved forecast accuracy across time horizons.
- Enhanced interpretability, enabling policy makers and agronomists to act on both climatic insights and yield projections.

Ultimately, these strategies collectively strengthen the scientific rigor and practical value of agricultural time series forecasting models.

## 5 Conclusion

In this project, we have developed a robust methodology for forecasting agricultural crop yields using ARIMAX time-series modeling. By integrating both auto-regressive components and relevant exogenous

variables from the most recent weeks, we were able to enhance the accuracy of yield predictions across multiple crops, states, and years. The systematic workflow - from data preprocessing, stationarity testing, and parameter tuning to final model evaluation - ensures the reliability and reproducibility of the results.

A key strength of this study lies in the inclusion of exogenous variables, which allowed the model to incorporate short-term influential factors that are often overlooked in traditional ARIMA models. Additionally, we compared model performance using both normalized and non-normalized data, highlighting the significance of feature scaling in improving predictive accuracy. The use of ADF and KPSS tests to determine the degree of differencing, and AIC/BIC for model selection, added statistical rigor to our model-building process.

The findings indicate that time-series modeling, when carefully calibrated and enhanced with auxiliary features, can serve as a valuable tool for agricultural yield forecasting. This is particularly important in data-scarce environments or regions where real-time monitoring and predictive planning are critical for food security and agricultural policy-making.

### 5.1 Limitations

Despite the promising results, several limitations of the current approach should be acknowledged:

- **Model Assumptions:** ARIMAX assumes linear relationships and Gaussian noise, which may not fully capture the complex, nonlinear, and often chaotic nature of agricultural systems influenced by numerous interdependent variables.
- **Limited External Variables:** Although we included recent historical characteristics as exogenous variables, broader environmental, economic and social factors - such as rainfall patterns, pest infestations, market fluctuations, and government interventions - were not considered.
- **Fixed Model Parameters:** The model parameters were tuned on historical data and kept fixed over time, which might limit adaptability to dynamic seasonal or climatic changes.
- **Spatial Generalization:** The model was trained and tested on his-

torical data from fixed locations. Its generalization ability to new regions or emerging crops has not been explored.

- **Computational Simplicity:** More advanced techniques such as dynamic regression, state-space models, or neural architectures were not applied, which might offer improved flexibility and accuracy.

## 5.2 Future Improvements

To address these challenges and build upon the foundation established in this study, future work may consider the following improvements:

- **Nonlinear Models:** Integrating machine learning algorithms such as Random Forests, Gradient Boosted Trees, or LSTM-based deep learning networks can capture nonlinearity and long-term dependencies more effectively.
- **Richer Feature Space:** Enhancing the model with high-resolution satellite data, real-time weather inputs, soil moisture, and vegetation indices can provide deeper insights into agro-climatic conditions.
- **Ensemble Techniques:** Combining multiple forecasting models could reduce individual model bias and improve robustness.
- **Region-Specific Calibration:** Tailoring models for specific geographical zones or crop types may increase precision and relevance in localized contexts.
- **Dynamic Updating:** Incorporating online learning or real-time model retraining pipelines would allow adaptation to recent trends and shocks.
- **Decision-Support Integration:** Linking predictive models with dashboards or mobile applications could translate forecasts into actionable insights for farmers, policymakers, and agribusiness stakeholders.

In conclusion, this work demonstrates the potential of statistical time-series models in improving agricultural intelligence and supporting proactive decision-making. With further refinement, such data-driven approaches can significantly contribute to sustainable agricultural planning, risk mitigation, and productivity enhancement.

# A APPENDICES

## A. List of Variables Used

- **Yield:** Crop yield value (dependent variable)
- **Exogenous Variables (last 12 weeks):**
  - Weekly Maximum Temperature
  - Weekly Minimum Temperature
  - Weekly Rainfall
  - Weekly Current Live Storage(FRL)
- **Date:** Daily timestamps
- **Crop Name, State:** Used for grouping and filtering

## B. Tools and Libraries Used

- **Python 3.10+**
- **Pandas, NumPy:** Data preprocessing and manipulation
- **Statsmodels:** Time-series modeling (ADF, KPSS, ARIMAX)
- **Matplotlib, Seaborn:** Data visualization
- **scikit-learn:** Data splitting, evaluation metrics

## C. Model Evaluation Metrics

- **Mean Absolute Error (MAE):** Measures average magnitude of errors
- **Root Mean Square Error (RMSE):** Penalizes larger errors more heavily
- **AIC/BIC:** Model selection criteria to balance fit and complexity

## D. ADF and KPSS Interpretation Guide

- **ADF Test:**
  - Null Hypothesis: Data has unit root (non-stationary)
  - Reject H0 if p-value < 0.05  $\Rightarrow$  Data is stationary
- **KPSS Test:**

- Null Hypothesis: Data is stationary
- Reject  $H_0$  if  $p\text{-value} < 0.05 \Rightarrow$  Data is non-stationary

## B References

### B.1 Video References

1. [Krish Naik Time Series Tutorial](#)
2. [Time Series Video Tutorial](#)

### B.2 Text References

1. Brockwell, P. J., & Davis, R. A. (2016). *Introduction to Time Series and Forecasting* (3rd ed.). Springer.
2. Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press.
3. Chatfield, C. (2004). *The Analysis of Time Series: An Introduction*. CRC press.